

CPE301 – SPRING 2024

Design Assignment 2

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Directory: https://github.com/JoshuaMa2003/submission_da

Video Playlist:

https://www.youtube.com/playlist?list=PLZ09MA_uFt63me60r5XWtDUj2m21n4nUP

1. COMPONENTS LIST

List of Components used

- Microchip Studio Debugger
- Microchip Studio Simulator
- ATmega328PB Microcontroller
- Saleae Logic Analyzer Software
- Logic Analyzer
- Male-to-Male Wire
- Female-to-Male Wire

2. INITIAL CODE OF TASK 2 (C)

```
/*
 * CPE301_DesignAssignment2_Task2_C.c
 *
 * Created: 3/9/2024 11:48:17 PM
 * Author : joshu
 */

#define F_CPU 16000000UL           //cpu clock speed at 16Mhz
#include <avr/io.h>
#include <util/delay.h>

int main(void){
    DDRB |= (1 << 5);              //configures PB5 as an output

    PORTB |= (1 << 5);             //sets PB5 to high

    DDRC &= ~(1 << 2);             //configure PC2 as an output

    PORTC |= (1 << 2);             //enable pull-resistor for PC2

    while(1){

        //check if PC2 is pressed
        //if PC2 is pressed, condition is true
        if(!(PINC & (1 << 2))){

            PORTB &= ~(1 << 5);     //set PB5 to low, turning LED off

            _delay_ms(2000);        //delay for 2 seconds
        }
        else
            PORTB |= (1 << 5);      //if the button is not pressed, set PB5 high
    }
    return 0;
}
```

2. INITIAL CODE OF TASK 3 (C)

```
/*
 * CPE301_DesignAssignment2_Task3_C.c
 *
 * Created: 3/9/2024 11:52:12 PM
 * Author : joshu
 */

#define F_CPU 16000000UL // Corrected CPU clock speed to 16MHz
#include <avr/io.h>
#include <avr/interrupt.h>
#include <util/delay.h>

int main(){
    DDRB |= (1 << 4) | (1 << 5); //set PB4 and PB5 as output pins
    DDRD &= ~(1 << 3); //set PD3 as an input pin
    PORTD |= 1 << 3; //activate pull-up resistor on PD3

    EICRA = 0x2; //configure INT1 to trigger on falling edge
    PORTB ^= (1 << 4) | (1 << 5); //toggle state of PB4 and PB5

    EIMSK |= (1 << INT1); //enable external interrupt INT1
    sei(); //enable global interrupts

    while(1){ //infinite loop
    }

ISR(INT1_vect){
    PORTB ^= (1 << 4); //toggle PB4 on interrupt

    _delay_ms(3000); //delay for 3000ms or 3s

    PORTB ^= (1 << 4); //toggle PB4 once again after delay
}
```

3. MODIFIED CODE OF TASK 2 (ASM) FROM TASK 2 (C)

```
;
; CPE301_DesignAssignment2_Task2_ASM.asm
;
; Created: 3/9/2024 11:37:48 PM
; Author : joshu
;

#define F_CPU 16000000UL
#include "m328pbdef.inc"

; Initialize
.org 0x00

CBI DDRC, 2          //set PC2 as an output
SBI DDRB, 5          //set PB5 as an input

BEGIN:
    LDI R16, 6        //loading loop counter for delay
    SBI PORTB, 5      //turn on the led at PB5
LOOP:
    SBIC PINC, 2      //wait for the button/switch on PC2
    RJMP LOOP        //stay in loop if button/switch not activated
    CBI PORTB, 5      //turn off the led at PB5

    C_DELAY:
    CALL DELAY        //call delay subroutine
    DEC R16           //decrement the loop counter
    BRNE C_DELAY      //repeat if it is not zero
    SBI PORTB, 5      //turn on the led at PB5 again

DELAY:
    LDI R18, 28       //delay loop: outer counter
    LDI R19, 8        //delay loop: middle counter
    LDI R20, 103      //delay loop: inner counter
L1:
    DEC R20           //decrement the inner counter
    BRNE L1          //continue inner counter if not zero
    DEC R19          //decrement middle counter
    BRNE L1          //continue middle loop if not zero
    DEC R18          //decrement outer counter
    BRNE L1          //continue outer loop if not zero
    NOP              //does nothing, only ensures delay timing

RET
```

3. MODIFIED CODE OF TASK 3 (ASM) FROM TASK 3 (C)

```
; Created: 3/9/2024 11:57:08 PM
; Author : joshu
;

#define F_CPU 16000000UL
#include "m328pbdef.inc"

.org 0x00 //begin at address 0x00
JMP BEGIN //jump to beginning of program
.org 0x04 //program counter for INT1
JMP ISR_INT1 //jump to ISR for INT1

BEGIN:
    LDI R16, (1 << 4) //load immediate value to set PB4 as output
    OUT DDRB, R16 //set DDRB to make PB4 an output

    LDI R17, (1 << 3) //load immediate value to enable pull-up on PD3
    OUT PORTD, R17 //enable pull-up for PD3
    LDI R18, (1 << ISC11) | (1 << ISC10) //INT1 to trigger on rising edge
    STS EICRA, R18 //external interrupt control register A

    LDI R19, (1 << INT1) //enable INT1
    STS EIMSK, R19 //external interrupt mask register

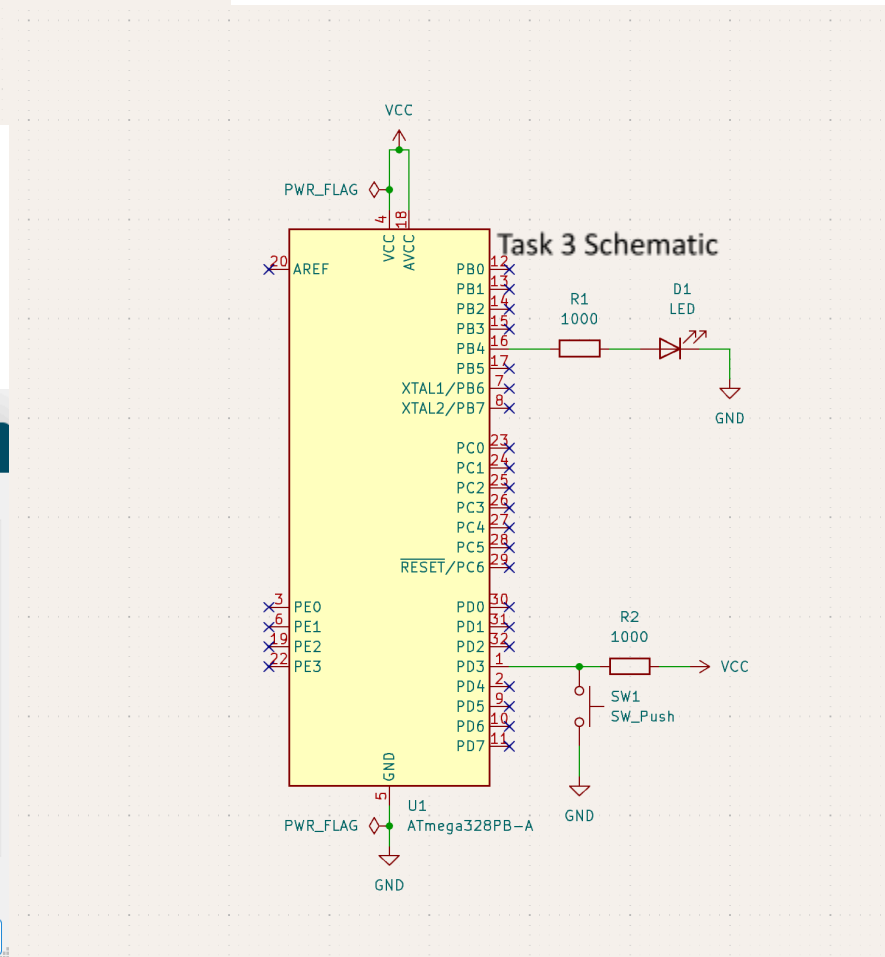
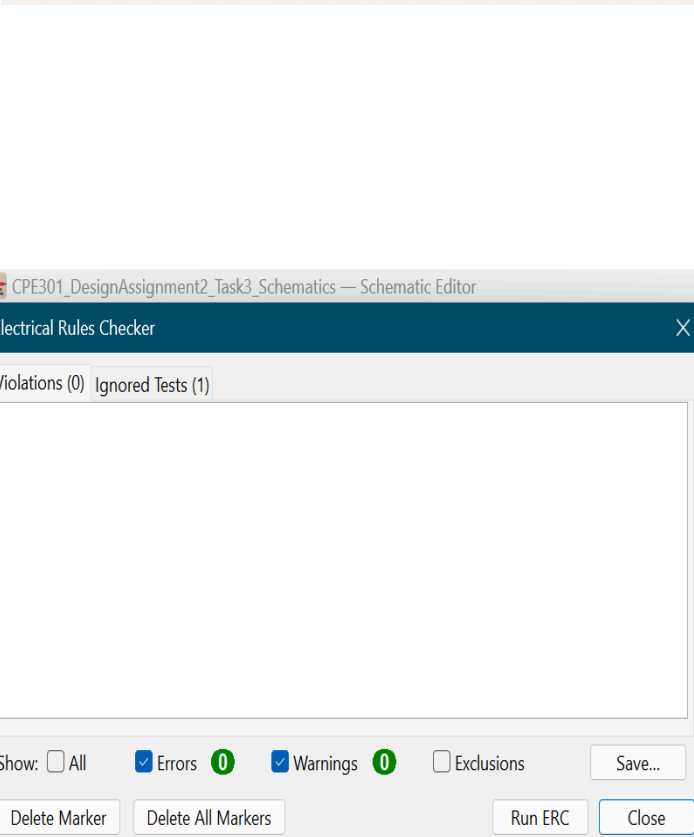
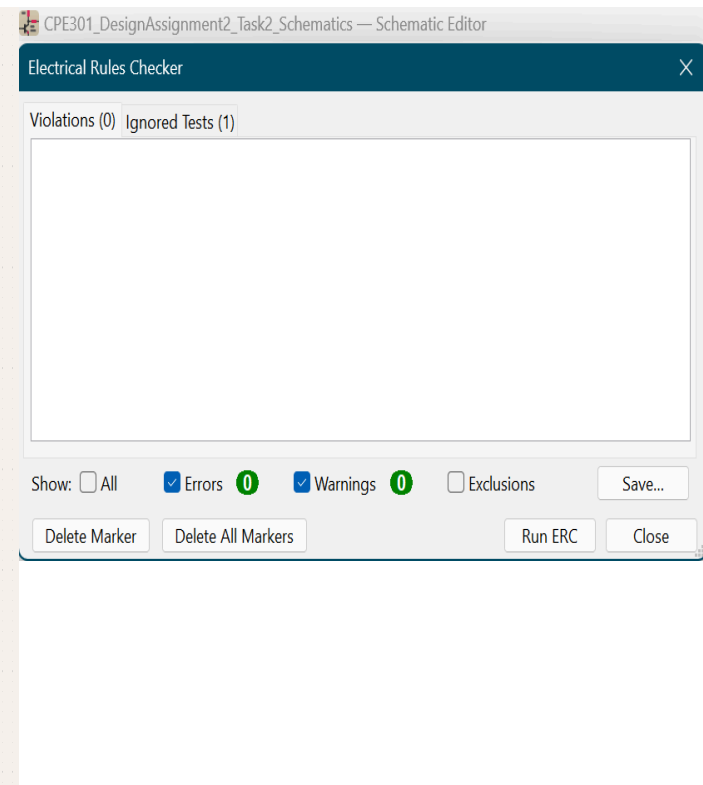
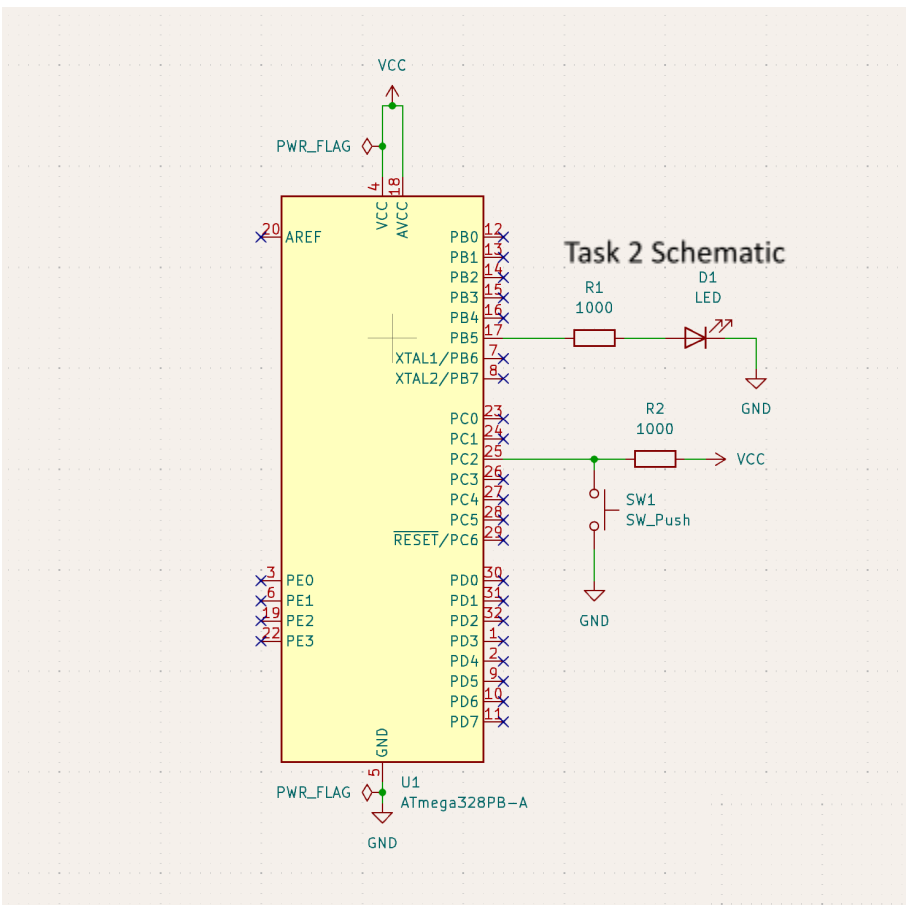
    SEI //enable global interrupts

LOOP:
    IN R20, PIND //read state of PORTD into R20
    ANDI R20, (1 << 3) //mask all but 3 bits
    CPI R20, (1 << 3) //compare masked value with expected bit
    BRNE ISR_INT1 //not equal, jump to ISR
    LDI R21, (1 << 4) //load value to set PB4 high
    OUT PORTB, R21 //set PB4 high
    RJMP LOOP //jump to loop

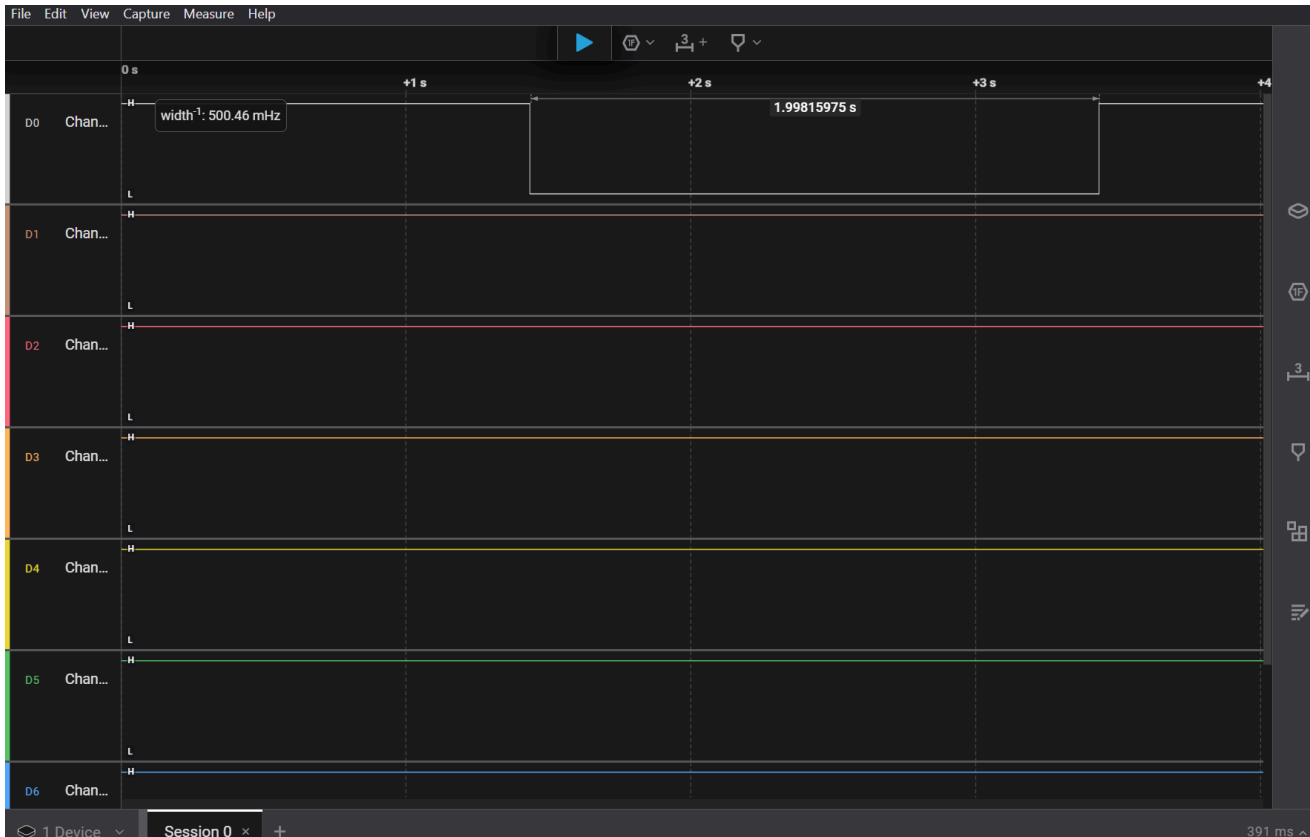
ISR_INT1:
    IN R22, PORTB //read current state of PORTB into R22
    ORI R22, ~(1 << 4) //clear bit 4 of PORTB
    ANDI R22, (1 << 5) //manipulate R22
    OUT PORTB, R22 //output value to PORTB
    CBI PORTB, 5 //clear bit 5 of PORTB
    RCALL DELAY //delay function
    RETI

DELAY:
    LDI R18, 255 //outer loop
    LDI R19, 255 //middle loop
    LDI R20, 23 //inner loop
L1: DEC R20 //decrement R20
    BRNE L1 //if R20, not zero, loop back
    DEC R19 //decrement R19
    BRNE L1 //if R19, not zero, loop back
    DEC R18 //decrement R18
    BRNE L1 //if R18, not zero, loop back
```

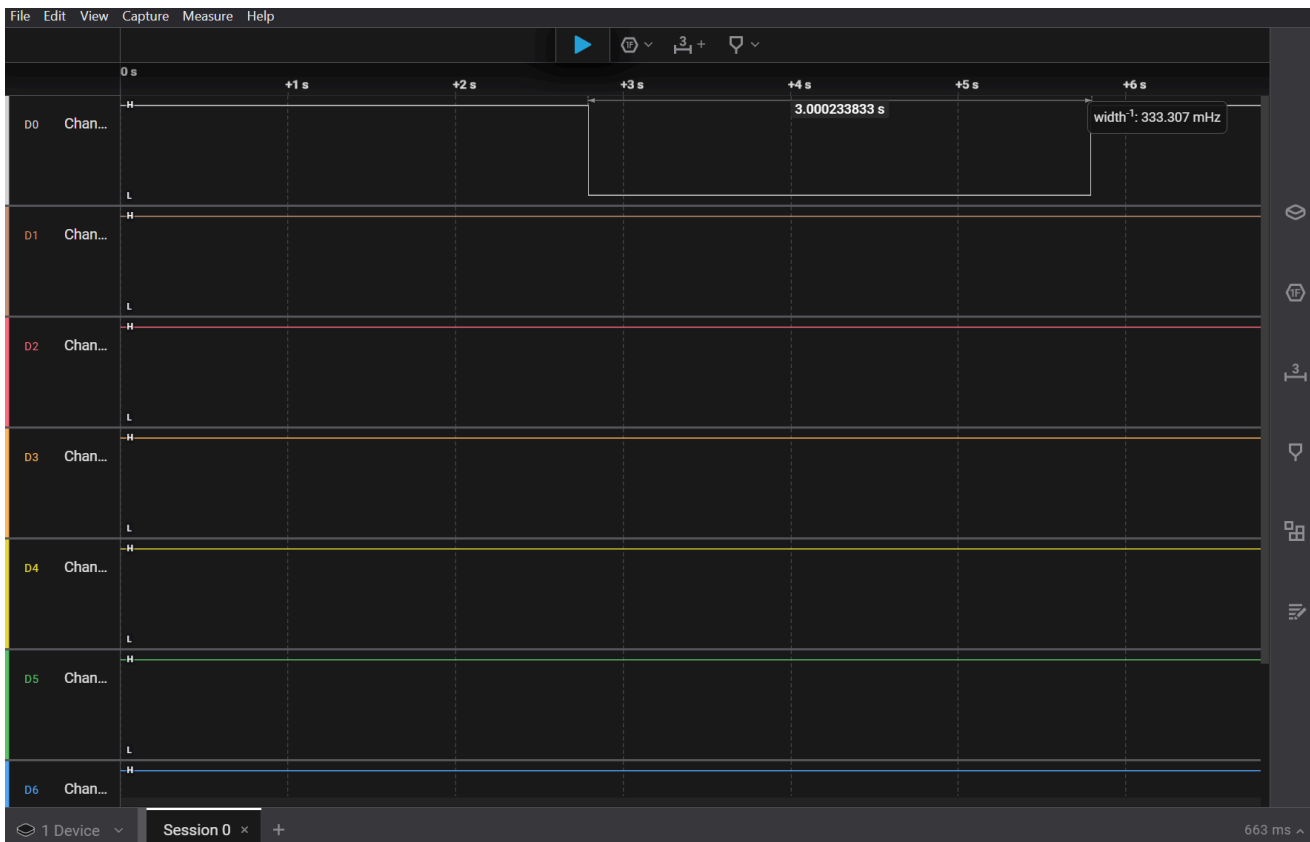
4. KICAD SCHEMATICS FOR TASKS 2 AND 3



5. SCREENSHOTS OF EACH TASK OUTPUT



Task 2 Output

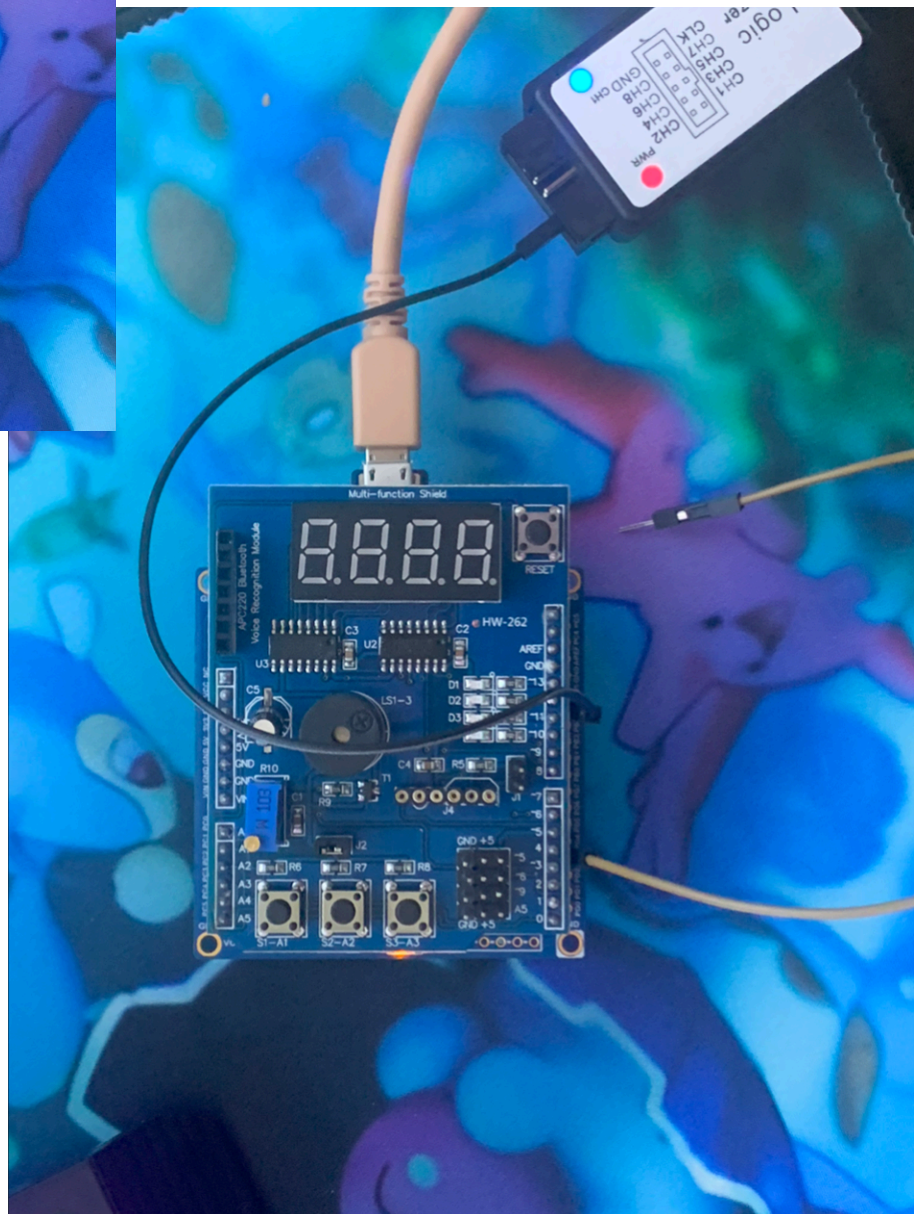


Task 3 Output

6. SCREENSHOTS OF EACH DEMO



Task 2 Demo



Task 3 Demo

7. VIDEO LINKS OF EACH DEMO

Task 2: https://youtu.be/1_UMxwOt0hM?feature=shared

Task 3: <https://youtu.be/XEJ1l3P38zg?feature=shared>

Student Academic Misconduct Policy

<http://studentconduct.unlv.edu/misconduct/policy.html>

"This assignment submission is my own, original work".

JOSHUA MARTINEZ